Piper of Barro Colorado Island

How Many Species Are on the Island and What Do We Know About Them?

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and M. Alejandra Jaramillo

FROM

The First 100 Years of Research on Barro Colorado:

Plant and Ecosystem Science, Volume 2

Smithsonian
Scholarly Press

WASHINGTON, D.C.
2024

Published by
SMITHSONIAN INSTITUTION SCHOLARLY PRESS
P.O. Box 37012, MRC 957
Washington, D.C. 20013-7012
https://scholarlypress.si.edu

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Recommended citation:
Lasso, E. and M. A. Jaramillo. 2024. Piper of Barro Colorado Island: How Many Species Are on the Island and What Do We Know About Them?. In *The First 100 Years of Research on Barro Colorado: Plant and Ecosystem Science*, Volume 2, ed. Muller-Landau, H. C. and S. J. Wright, pp. 691–700. Washington, DC: Smithsonian Institution Scholarly Press. https://doi.org/10.5479/si.26880856

ISBN (online, 2 vols.): 978-1-944466-70-1 ISBN (print, 2 vols.): 978-1-944466-71-8

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Piper of Barro Colorado Island

How Many Species Are on the Island and What Do We Know About Them?

Eloisa Lasso[1]* *and M. Alejandra Jaramillo*[2]

ABSTRACT. *Piper* is one of the most diverse and abundant genera at Barro Colorado Island (BCI) and a fascinating model system given its diversity in habitats and ecological and physiological strategies. In this chapter, we review and summarize what is known about the life history of *Piper*. We begin by revising and updating the list of species on BCI to 27 species. We discuss the hypotheses proposed to explain their coexistence. We reflect on the ability of many *Piper* species to spread clonally and explore the potential causes and consequences of such propagation. We review current knowledge on *Piper*'s pollination and reproductive biology and discuss how its pollination, seed dispersal, and clonal spread affect gene flow and population genetic structure. We conclude by discussing the reasons for the decline of BCI *Piper* populations. We also identify several open research questions, hoping to inspire future generations to continue using *Piper* as a model system.

Keywords: breeding system; clonal spread; gene flow; model system; *Piper* taxonomy; pollination biology; population decline; population structure; species coexistence; 50-ha plot

INTRODUCTION

Piper is one of the five most speciose genera in the Neotropics (Gentry, 1990; Draper et al., 2021), containing about 2,600 species worldwide and 736 species in Mesoamerica, of which 606 are endemic (Callejas-Posada, 2020). This highly diverse genus with a common evolutionary heritage and with species across the full spectrum of tropical rainforest functional diversity (e.g., shade tolerant, light demanding, associated with ravines, from lowland forests to montane cloud forests) provides a perfect model system for ecological studies and to assess the functional role of plant traits (Field and Vázquez-Yanes, 1993). Recognizing the promising potential of *Piper* as a model system, Dyer and Palmer (2004), compiled all that was known about *Piper*'s biology and ecology. Eighteen years have passed since the publication of their book, and although in this chapter we do not provide an exhaustive review of advances in *Piper* research, we do summarize research done on *Piper* in the Barro Colorado Nature Monument (BCNM) and place it in the context of research done elsewhere.

Researchers interested in *Piper* have accepted 22 species on Barro Colorado Island since the publication of the *Flora of Barro Colorado Island* and the *Seedlings of Barro*

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Manuscript received 14 November 2022; accepted 17 July 2023.

TABLE 1. *Piper* species reported at Barro Colorado Island (BCI) with currently accepted names in Flora Mesoamericana (Callejas-Posada, 2020) and the name used in *The Flora of Barro Colorado Island* (Croat, 1978). Habitat is from herbarium sheets, Callejas-Posada (2020), and personal experience. Reference specimen refers to one specimen deposited in the Missouri Botanical Garden (MO) with the collector's name and year. References refer to where the species is recorded as present at BCI.

Current name	Name in Croat (1978)	Habitat	Reference specimen	References[a]
Piper albopunctulatissimum Trel.	*Piper imperiale*	Forest, creek bank	T. B. Croat 12195, 1970	1, 2
Piper collium (Trel.) Callejas	*Piper aequale*	Forest and gaps	T. B. Croat 11849, 1970	1, 3
Piper amphioxys Trel.	*Piper aequale*	Forest, creek bank	T. B. Croat 11938, 1970	1
Piper auritum Kunth	*Piper auritum*	Clearings and gaps	T. B. Croat 12429, 1970	1, 2, 3
Piper cabagranum C.DC.	*Piper aequale*	Forest	T. B. Croat 10096, 1970	1
Piper casimirianum Hemsl.	*Piper carrilloanum*[b]	Forest, creek bank	T. B. Croat 12212, 1970	1, 2, 3
Piper cordulatum C.DC.	*Piper cordulatum*	Forest and gaps	T. B. Croat 7096, 1969	1, 2, 3
Piper culebranum C.DC.	*Piper culebranum*	Forest, creek bank	T. B. Croat 10221, 1970	1, 2, 3
Piper diazanum Trel.	*Piper hispidum*	Forest, trails, creek bank	T. B. Croat 8590, 1969	1, 2, 3
Piper darienense C.DC.	*Piper darienense*	Forest	T. B. Croat 12210, 1970	1, 3
Piper euryphyllum C.DC.	Not recorded	Forest	*P. longispicum* sensu Condit, no ref. spec.	2
Piper grande Vahl	*Piper grande*	Forest, creek bank	T. B. Croat 7099, 1969	1, 2, 3
Piper leptocladum C.DC.	*Piper dilatatum*	Lake shore, clearings	T. B. Croat 12263, 1970	1
Piper lucigaudens C.DC.	Not recorded	Forest, creek bank	J. E. Ebinger 60, 1960	1
Piper marginatum Jacq.	*Piper marginatum*	Clearings	T. B. Croat 12875, 1970	1, 2, 3
Piper multiplinervium C.DC.	*Piper aristolochiifolium*	Forest, creek bank	T. B. Croat 12929, 1970	1, 3
Piper peltatum L.	*Pothomorphe peltate*	Clearings	T. B. Croat 5953, 1968	1, 3
Piper perlasense Yunck.	*Piper perlasense*	Forest, Lake shore	T. B. Croat 4984, 1968	1, 3
Piper playa-blancanum Trel.	*Piper aequale*	Forest	T. B. Croat 10136, 1970	1, 2
Piper pseudogaragaranum Trel.	*P. pseudogaragaranum*	Forest, creek bank	T. B. Croat 10829, 1970	1, 3
Piper pseudovariabile Trel.	Not recorded	Forest, creek bank	T. B. Croat 11259, 1970	1
Piper pubistipulum C.DC.	*Piper pubistipulum*	Forest, creek bank	T. B. Croat 10884, 1970	1, 3
Piper reticulatum L.	*Piper reticulatum*	Clearings, gaps, forest	T. B. Croat 14651, 1971	1, 2
Piper seducentifolium Trel.	*Piper arieianum*	Forest, clearings	T. B. Croat 13249, 1971	1
Piper subnudispicum Trel.	*Piper arboreum*	Forest, creek bank	T. B. Croat 12605, 1970	1, 3
Piper villiramulum C.DC.	*Piper villiramulum*	Clearings	T. B. Croat 12812, 1970	1, 3
Piper viridicaule Trel.	*Piper viridicaule*	Forest, creek bank	P. C. Standley 31402, 1924	1, 3

[a] References: 1. Symbiota (2022); 2. Condit et al. (2019); 3. Croat (1978).

[b] Previously identified as *P. schiedeanum* in several specimens.

Colorado Island and the Neotropics (Croat, 1978; Garwood and Tebbs, 2009). Here we revise and update the list of *Piper* species for BCI using the most recent revisions of the genus in Flora Mesoamericana (Callejas-Posada, 2020) and the newest determinations of specimens deposited in the Missouri Botanical Garden by Croat and others, which now are available in the STRI Symbiota portal (Symbiota, 2022). Following this revision, 27 *Piper* species have been recorded for BCI (Table 1). All 27 species are confirmed with samples collected on BCI. One additional species (*P. peracuminatum*) was reported by Standley without a specimen, has not been encountered on BCI since Standley (Croat, 1978), and is omitted here. Of the six species newly recorded since 1978, one was recorded in the 1985 census of the 50-ha plot and five were collected by 1970 and have

new determinations by Callejas-Posada. Not one of the species "new" to BCI was detected in previously underexplored areas or is a recent colonist.

SO MANY *PIPER* SPECIES AT BCI; HOW DO THEY COEXIST?

With 27 *Piper* species coexisting within BCI and occupying such diverse habitats including understory, clearings, and ravines (Table 1), we have an ideal model for studies on function and ecology, especially because BCI hosts species from most major clades of Neotropical *Piper* (Jaramillo et al., 2008; Fig. 1). It is impossible not to encounter at least one species of *Piper* when walking in the BCI forest. They are generally easy to identify, with their thickened stem nodes and spike-like white inflorescences, full of tiny, sessile, hermaphroditic flowers (Fig. 2c), or with their green infructescences ripening in the late afternoon and evening and giving off a sweet smell that attracts their most frequent dispersers, the *Carollia* bats (Thies and Kalko, 2004; Santana et al., 2021). *Piper* is abundant in the shrub layer and plays a key role in supporting the frugivorous bat community at BCI and other forests (Fleming, 1985; Thies and Kalko, 2004).

The co-occurrence of so many *Piper* species raises the classic ecological question: how do they manage to coexist? One possibility is that coexistence occurs because different species occupy different ecological niches. *Piper* differ by occupying various habitats, from well-illuminated treefall gaps to deeply shaded forest understories. Diversity in growth forms and size, including short to tall shrubs, climbing shrubs, and treelets, have been proposed as key to partitioning light niches in the forest understory (Marquis, 2004). Some BCI species grow only associated with slopes or streams (Harms et al., 2001), which probably indicates their water requirements, whereas others prefer

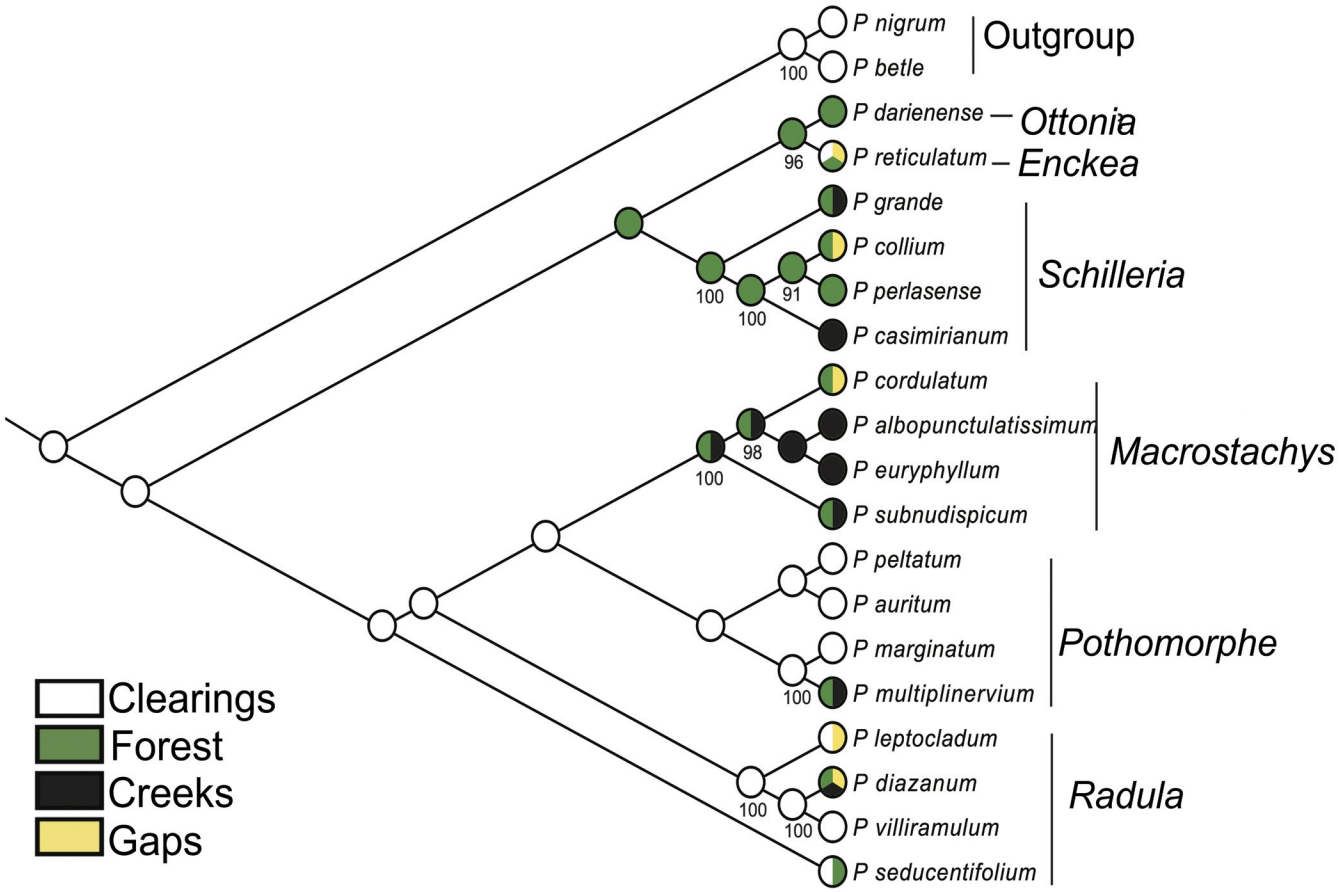

FIGURE 1. Maximum likelihood reconstruction of preferred habitat for *Piper* species occurring on Barro Colorado Island (except nine taxa for which sequences are not available). Maximum likelihood inference of phylogeny based on internal transcribed sequences (ITS). Clade (subgenera) names are written on the right. The outgroup are *Piper* species from Asia. The areas in the pie diagrams represent the relative support for each reconstructed habitat.

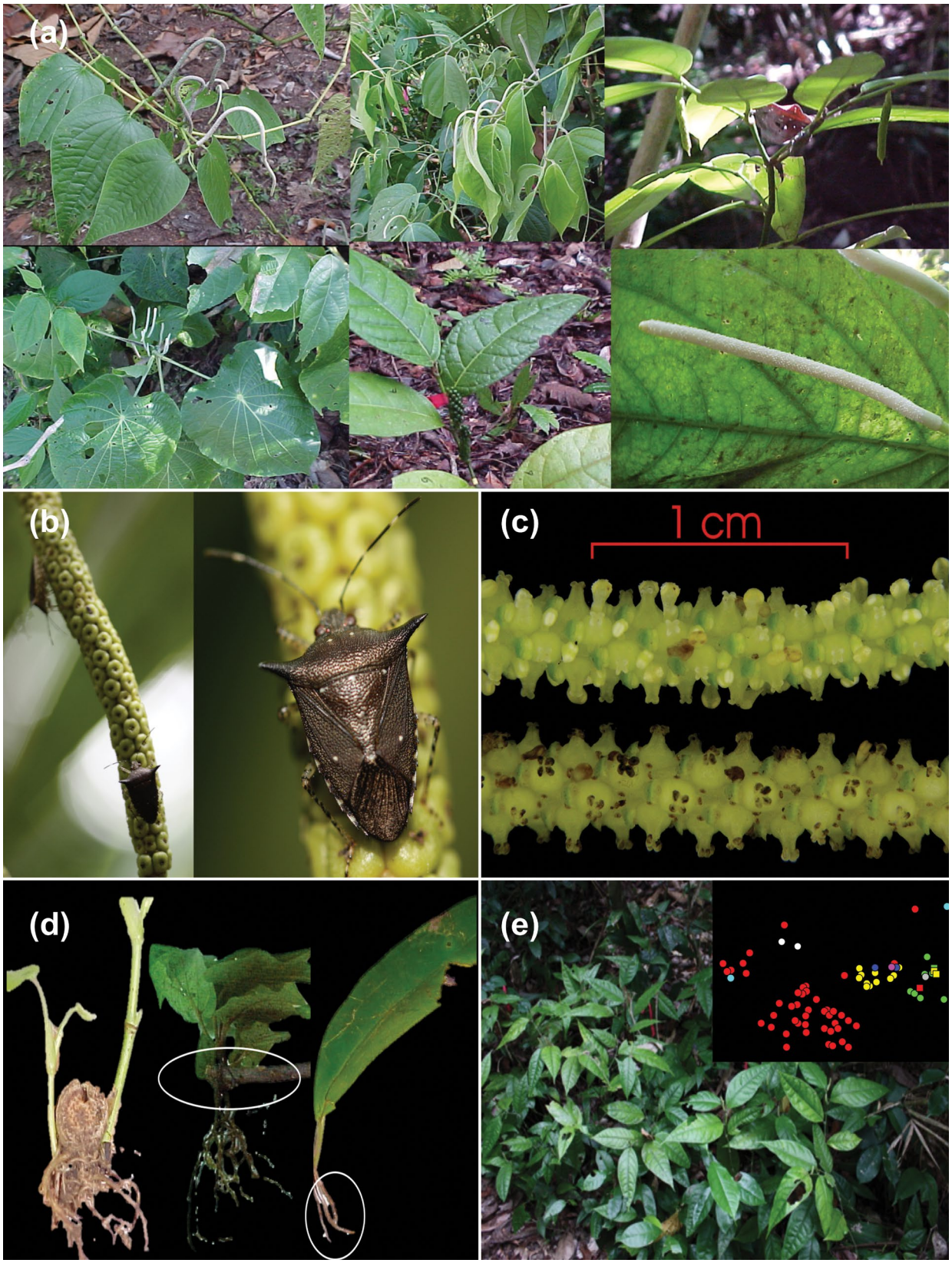

FIGURE 2. Diversity of *Piper* species at Barro Colorado Island (BCI). (**a**) Diversity of *Piper* species with shared characteristics, such as thickened nodes and white spike-like inflorescences. (**b, left and right**) *Sibaria englemanim*, a *Piper* pre-dispersal seed predator at BCI. (**c**) Inflorescences of *Piper darienense* with hundreds of tiny hermaphroditic flowers. (Photo by Steve Paton, courtesy of Smithsonian Institution.) (**d**) *Piper* producing new roots and shoots (circled) that often result in the recruitment of new clonal plants. (**e**) Aggregations of *Piper darienensis* on a BCI trail. Each "individual" has just one to perhaps six leaves. In the inset figure, which has an area of 600 m^2, each color represents a unique genotype identified with genetic markers where the red genotype is one large clone (Lasso et al., 2012).

drier sites (Engelbrecht et al., 2000). Differences in germination and seedling establishment requirements in species from different habitats allow each species to germinate in suitable microsites for seedling establishment and growth, thereby reducing interspecific competition and contributing to coexistence (Daws et al., 2002; Lasso et al., 2009).

Piper species also differ during the reproductive and regeneration phase (Fleming, 1985; Daws et al., 2002; Lasso et al., 2009). Differences among species in the timing of flowering and fruiting allow them to coexist despite sharing the same pollinators and seed dispersers, minimizing hybridization and competition for animal visitors (Fleming, 1985; Thies and Kalko, 2004).

Herbivores, through apparent competition, might also promote the coexistence of *Piper* on BCI (e.g., Marquis, 1988). At La Selva, Costa Rica, diverse clumps of *Piper* suffered less herbivory than less diverse clumps, and dispersal by generalist bats like *Carollia perspicillata* was fundamental in creating these diverse *Piper* patches (Salazar et al., 2013). Chemical differences among *Piper* species have also been proposed to be fundamental for coexistence of this taxon (Salazar et al., 2016). The diversity of secondary metabolites was greater for *Piper* and other species-rich genera from BCI than it was for 69 phylogenetically diverse species from a Northeastern deciduous forest in North America (Sedio et al., 2018). In addition, chemical compositional similarity was uniformly low for congeners from BCI. Both observations are consistent with the hypothesis that stronger plant-enemy interactions lead to more rapid divergence and greater metabolomic variation in tropical than in temperate plants (Sedio et al., 2018).

An additional axis of differentiation for *Piper* includes species resprouting capacity (Lasso et al., 2009). By including resprouting capacity (the persistence niche) with sexual traits (the regeneration niche) and a suite of functional traits, we identified three life-history strategies at BCI (Lasso et al., 2009). First, *a Pioneer species* group that includes species that live exclusively in gaps, use resources to grow to maturity rapidly, and produce many tiny seeds which presumably allows them to colonize gaps. These species are largely incapable of reproducing asexually. This group includes *P. marginatum*, *P. peltatum*, and *P. leptocladum*. The second life-history strategy, *a cryptic gap-dependent group*, includes species that thrive in gaps but that are also tolerant of understory light conditions as juveniles and adults. They also produce many tiny seeds but have an intermediate capacity to reproduce asexually, allowing them to persist and spread to some degree in the understory. This group includes *Piper casimirianum*, *P. collium*, and *P. grande*. The third life-history strategy, *a shade-tolerant group*, includes species with low seed output but with higher capacity to resprout and spread asexually, allowing them to persist and spread in the understory once they have established. This group includes *P. darienensis* and *P. cordulatum*. This last group still invests heavily in fruits and seed production, so bat dispersal might favor colonization of new sites followed by population expansion through asexual reproduction. To classify the remaining BCI *Piper* species into these groups, more systematic data on seed size, germination success, and asexual reproductive ability are needed. The shade-tolerant species are found in the clade Ottonia (*P. darienense*) and Macrostachys (*P. cordulatum*), the cryptic gap-dependent species are found in the Schilleria clade, and the light-demanding pioneers are found in the Pothomorphe (*P. marginatum, P. peltatum*) and Radula clades (*P. leptocladum*; Fig. 1).

WHY DOES *PIPER* PROPAGATE ASEXUALLY AND FORM CLONAL AGGREGATIONS?

Piper species often have trouble recruiting sexually, especially understory species. The first major filter is high predispersal seed predation. Up to 76% of seeds may be lost to predators in the understory, although these numbers go down to 12% in gaps (Greig, 1993a). Species of Hemiptera are commonly observed sticking their stylet into immature seeds (Fig. 1b). Post-dispersal seed predation can also be high; around 75% of seeds of *P. amalago* were consumed in a dry forest in Costa Rica (Perry and Fleming, 1980). Seeds that survive predation have exceptionally low emergence success. Most of the tiny (<1 mg) photoblastic seeds (Orozco-Segovia and Vázquez-Yanes, 1989; Daws et al., 2002) fail to emerge when sown in the understory. Only 0.2% of 8000 seeds sown in the understory emerged for five *Piper* species studied at BCI (Lasso et al., 2009). Likewise, Fleming (1981) found only four surviving seedlings of 1,600 seeds of *Piper amalago*, and Greig (1993a) observed only five surviving seedlings from 3,000 seeds of five *Piper* species, experimentally sown into both gap and understory conditions. With this strong seedling establishment limitation, some form of asexual reproduction seems critical for *Piper* population persistence in the understory (Lasso et al., 2012), and shade-tolerant species are particularly successful at resprouting (Greig, 1993b; Lasso et al., 2009).

Two main methods of asexual reproduction have been observed at BCI; layering and fragmentation (Fig. 2d). In layering, new plants originate at nodes from branches and trunks that are pinned to the ground by branches and litterfall. The original shoot may eventually rot away, leaving independent ramets. In fragmentation, new plants sprout from stem or leaf tissue portions that remain alive on moist soil or leaf litter layers (Greig, 1993b; Lasso et al., 2009). In both cases, physiologically independent ramets are produced that are difficult to identify as clones unless molecular markers are used (Lasso, 2008). Using molecular markers to identify clones, we found that shade-tolerant *Piper* species from BCI recruited asexually more frequently than light-demanding species (Lasso et al., 2012). In understory plant aggregations, 36% to 42% of sampled shoots were clones (Fig. 2e). In contrast, clones were absent in most cases for light-demanding species, and only for one clump of *P. marginatum* did we find that 26% of shoots were clones (Lasso et al., 2012).

One consequence of asexual reproduction, which has both benefits and costs, is that offspring are genetically identical to

each other and their parent. Such genetic similarity may be advantageous for the population if their genotypes are well adapted to the environment but disadvantageous if climate changes, reducing the population's long-term resilience to environmental change (Hamilton, 1980; Burt, 2000). It is also possible that *Piper* species are risking their future sexual success by reproducing clonally too often. The larger the clones are, the more likely self-pollination and inbreeding will occur, as we will explain.

WHO POLLINATES *PIPER* AT BCI?

Little is known about the pollination biology of *Piper* species on BCI. The most detailed study, done on *Piper leptocladum* (formerly *P. dilatatum*), found a diverse assemblage of insects visiting inflorescences, including bees of the genus *Megachile* (Megachilidae), stingless bees (Apidae), and syrphid flies (Kikuchi et al., 2007), all documented *Piper* pollinators elsewhere (Semple, 1974; Fleming, 1985; De-Figueiredo and Sazima, 2000; Valentin-Silva et al., 2021). The most common visitors were *Trigona* species, *T. dorsalis*, and *T. muzoensis*, which were responsible for 66% of all visits. We have no evidence that any *Piper* species on BCI are wind pollinated. This is still possible, however, given reports of anemophily for *Piper* species in Brazil (De-Figueiredo and Sazima, 2000). Most insect visitors visited both shade-tolerant and light-demanding species of *Piper* at BCI, corroborating observations that *Piper* species from different habits share pollinators, as shown by Fleming (1985) in Costa Rica. From our observations over several years, we cannot rule out the possibility of hybridization between BCI *Piper* species. We have seen both an overlap in flowering times between species and the same insect species visiting multiple species simultaneously. We have also seen some *Piper* individuals with intermediate forms between two species that could be hybrids. This is a fascinating topic that hopefully someone will explore in the future.

HOW FAR DO THE POLLEN AND SEEDS MOVE, AND WHAT DOES THIS MEAN FOR *PIPER*?

Observation of the behavior of *Piper* inflorescence visitors indicates that its pollinators are likely to promote self-pollination. Insect visitors move mainly up and down the same inflorescences. Several inflorescences in the same plant are visited before the visitor leaves, and only occasionally do they move between neighboring plants (Semple, 1974; Kikuchi et al., 2007). Our calculations of how far pollen can move using fluorescent powder as a pollen analog suggest that the farthest pollen movement between flowers is 340 m with a very low probability of occurrence (Kikuchi et al., 2007). Using microsatellites and paternity analysis, we calculated that pollen moves 20 m in *P. cordulatum* (unpublished data). Self-pollination seems to be a common feature in *Piper* (Valentin-Silva et al., 2021) and can lead to self-fertilization if species are self-compatible, as many *Piper* species are (Marquis, 1988; De-Figueiredo and Sazima, 2000; Kikuchi et al., 2007).

In a series of pollinator exclusion and hand-pollination experiments with self- and cross-pollen, we found that five of eight *Piper* species at BCI had self-compatibility indices (SCI) greater than 0.75 (Lasso, unpublished). SCI is the ratio of the fruit set of self-pollinated to cross-pollinated inflorescences, where fruit set is the number of infructescences produced from the number of inflorescences self- or cross-pollinated. SCI values greater than 0.75 are indicative of self-compatibility (Lloyd and Schoen, 1992). Two of the remaining species had intermediate values of SCI (0.56 and 0.69), indicating that some selfing is occurring, and only one species, *Piper marginatum*, was definitively not self-compatible (Lasso, unpublished). In other words, we have a series of species that can multiply clonally, that are visited by pollinators that move pollen short distances, and that are self-compatible. All this could lead to the possibility that the low germination and establishment success observed to date result in part from inbreeding depression, a possibility that should be evaluated by studying the viability of seeds derived from self- and cross-pollination.

Studies using allozymes and amplified fragment-length polymorphisms (AFLP) to evaluate spatial genetic structure in *Piper* support this view (Heywood and Fleming, 1986; Mariot et al., 2002; Lasso et al., 2011). The reproductive success of *Piper* is clearly affected by who their neighbors are. In our latest study, we identified solitary plants of *P. cordulatum* and plants surrounded by different numbers of clones using molecular markers and observed their reproductive success. We found that plants with identical neighbors do not do as well in their reproductive success (Fig. 3) as other plants with genetically distinct neighbors. *Piper* may be stuck in a cycle in which self-compatibility and the ability to reproduce asexually are reducing the ability to reproduce sexually.

Bats are the main seed dispersers of *Piper* and may help maintain gene flow. Bat dispersal also increases windows of opportunity for colonization events while reinforcing high population structure (creating subgroups of related plants). Bats can transport *Piper* seeds within a range of 50 to 300 m and occasionally over longer distances of >1 km (Fleming, 1981; Thies, 1998). But they also transport whole infructescences to feeding roosts (Fleming and Heithaus, 1981), tending to disperse closely related seeds in clumps. Genetic spatial autocorrelation analysis indicates that siblings are recruiting together (Lasso et al., 2011). Interestingly, not all plant aggregations have the same origin; sometimes plants in a clump are siblings, sometimes they are clones, and some aggregations were composed of unrelated plants (Lasso et al., 2011). The combined effects of limited seed and pollen dispersal, clonal spread, and selfing may have facilitated the diversification of populations in response to local natural selection or genetic drift, leading eventually to speciation, which could explain the large number of endemics and the remarkable diversity of this rich genus.

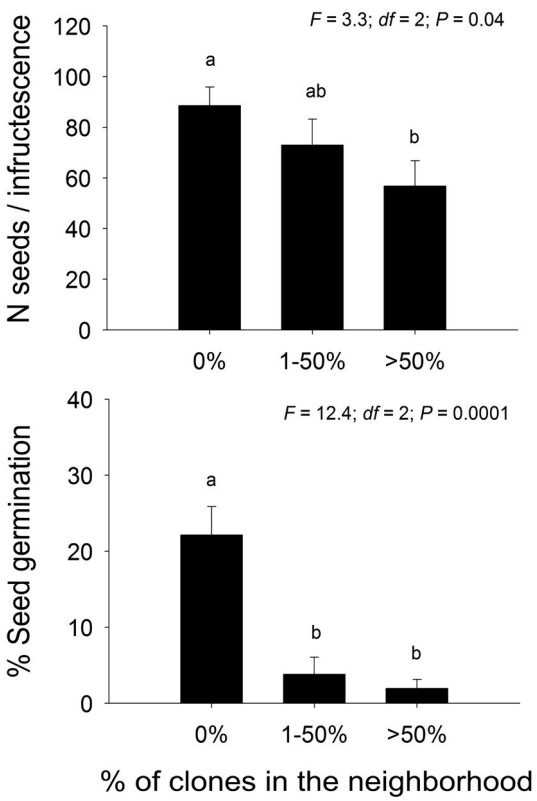

FIGURE 3. Effect of the presence of neighboring clones on the reproductive success of *Piper cordulatum*. Statistically significant differences in reproductive success do not share lower-case letters (from an ANOVA and Tukey test). The top panel shows the number of full-grown viable seeds on each infructescence. Fruits and seeds were collected from 39 maternal plants scattered across the landscape on Barro Colorado Island. Nineteen adult plants had no clonal neighbors (0% clone) within a radius of 20 m, 10 plants had 1% to 50% of clonal neighbors, and 10 plants were surrounded by more than 50% clonal neighbors. From each plant, we collected three to six infructescences. From each infructescence, we extracted 30–172 healthy (no signs of predation) seeds. A total of 184 infructescences and 13,613 seeds are represented in the top and bottom panels, respectively.

PIPER DECLINE ON BCI: WHY?

Populations of several *Piper* species have declined over time in the long-term temporal data from the BCI 50-ha plot (Condit et al., 2019; Fig. 4). The most impressive case is *Piper cordulatum*, which had 4,650 individuals recorded during the first census conducted in 1982 and just 98 individuals recorded 33 years later in 2015. Other *Piper* species have been declining, although it is not clear why. One hypothesis proposed to explain this population decline is that *Piper* species are sensitive to El Niño–associated drought and to a long-term drying trend in central Panama (Condit et al., 1996). Consistent with this hypothesis, the decline started after the long and dry el Niño event in 1983. *Piper* populations have not recovered, however, and few recruits were recorded in the 2010 and 2015 censuses of the 50-ha plot even though regional rainfall returned to high levels last seen in the 1930s between 1980 and 2015 (Paton and Stallard, in press).

Piper species from BCI are probably xerophobic. Most species in this genus prevail in wet and cloudy tropical forests, like other early diverging angiosperms (Field et al., 2009; Pouteau et al., 2020). *Piper* xylem vessels have scalariform perforation (Callejas-Posada, 1986), and low leaf vein density (Boyce et al., 2009; Field and Bodribb, 2013), characteristics that lower hydraulic efficiency (Sperry et al., 2007). For this reason, it is not surprising that many *Piper* species are highly susceptible to drought (as are other Magnoliidae; Engelbrecht and Kursar, 2003). It is possible that, in addition to low tolerance to drought affecting adults and the recruitment of seedlings (Engelbrecht and Kursar, 2003), *Piper* faces problems of sexual recruitment, as various studies have shown (Greig, 1993a; Daws et al., 2002; Lasso et al., 2012). On the one hand, we have populations hit by several episodes of drought, exacerbated by climate change. On the other hand, the biology of the genus provides limited escape mechanisms, with elevated inbreeding and low reproductive success possibly hindering recovery from drought. We are waiting with some apprehension for the results of the next census when we will see the effect of the last 2016 El Niño event to understand whether this 33-year decline continues. If the trend continues, we need urgent research to understand whether the declines observed in the 50-ha plot are occurring regionally and to identify the specific causes of the decline.

We do not, however, want to end this chapter on such a depressing note. We believe that the prospect is not that gloomy as all *Piper* populations continue to maintain relatively high genetic diversity (Lasso et al., 2012), as do other clonal plant species (Ellstrand and Roose, 1987; Silvertown, 2008; Karlin et al., 2012; Huang et al., 2021). One big question that remains to be answered is how genetic diversity is generated and whether it can be sustained in the long term. It is possible that some of the genetic variation arises from somatic mutations that pass into reproductive tissues (Schoen and Schultz, 2019). It is also possible that solitary *Piper* individuals or those in genetically diverse clumps reduce population-level inbreeding and contribute to genetic admixture. Perhaps, sporadic windows of opportunity for sexual recruitment are sufficient to maintain high genetic diversity. Finally, it cannot be ruled out that frequency-dependent mortality could maintain genetic diversity by eliminating higher frequency clones. Most likely, it is a combination of several of these. We are trying to understand some of these processes through a long-term experiment following artificially created clonal populations of

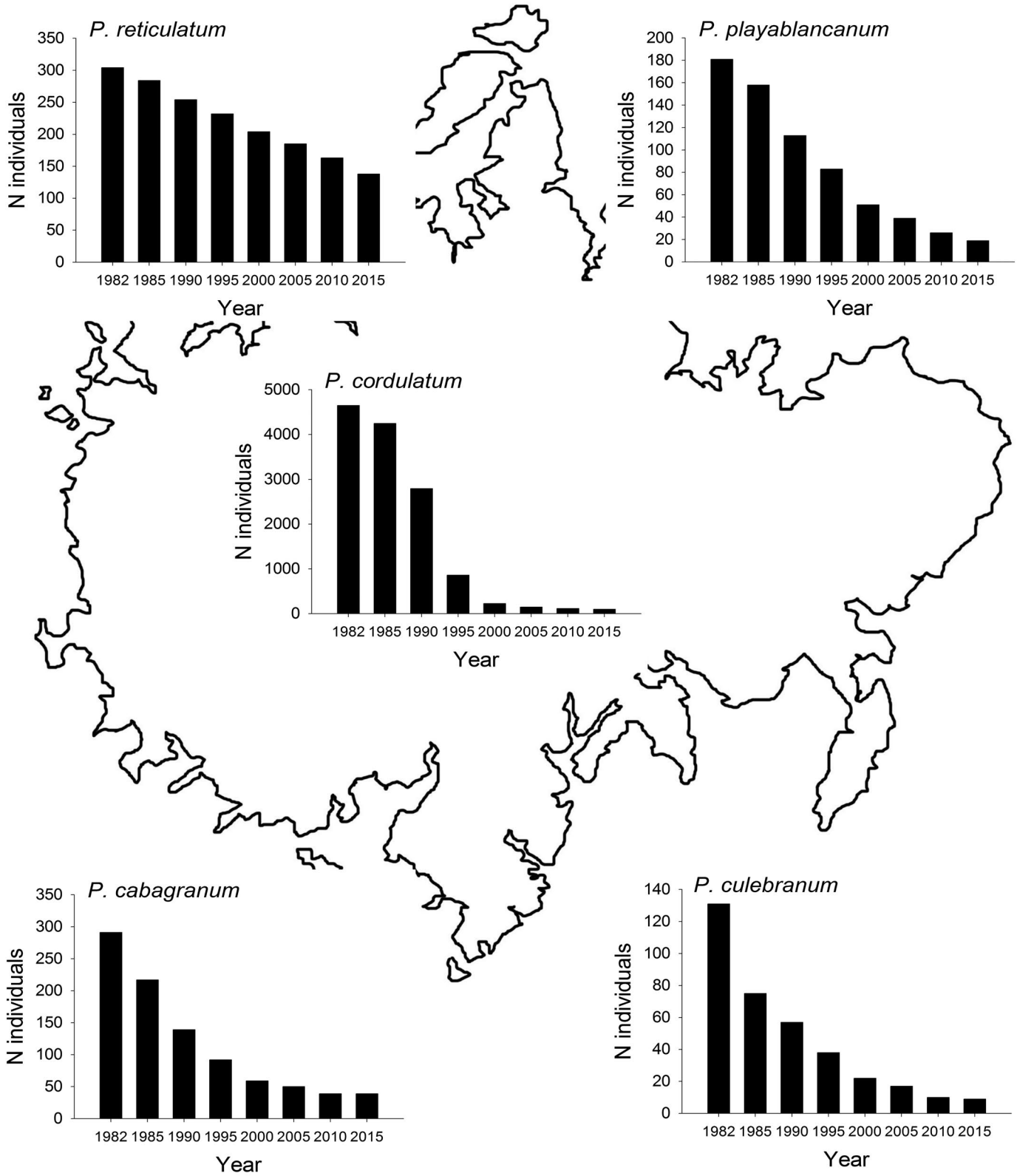

FIGURE 4. *Piper* decline on Barro Colorado Island. Each bar graph shows the number of live individuals during each census for five of the most common *Piper* species in the 50-ha plot starting in 1982. Graphs were created with census data from Condit et al. (2019).

P. cordulatum on the Buena Vista Peninsula in the BCNM. In 2023, the experiment celebrates its 10th anniversary as BCI celebrates its 100th.

ACKNOWLEDGMENTS

Special thanks to all the BCI staff who always assisted in all logistical aspects of the various projects undertaken to understand BCI's *Piper*. Thanks to Joe Wright, Jim Dalling, and Bob Marquis for reading this chapter and for providing thoughtful comments and suggestions for its improvement.

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